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Unraveling the underlying structural & transport mechanism of lithium-ion within Lithium bis(trifluoromethanesulfonyl)imide subjected to organic & inorganic matrix based Eutectogel

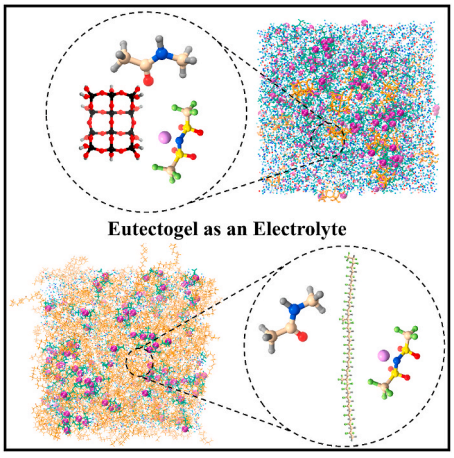
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HIGHLIGHTS

- Molecular dynamics explore eutectogel system's structural and transport properties
- Eutectogel made of DES(LiTFSI:NMAC = 1:4) in polymers(SiO_2/PVDF-HFP)
- The structural properties are calculated using RDF and H-bond
- The transport properties are examined using MSD and ionic conductivity

GRAPHICAL ABSTRACT



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ABSTRACT

Eutectogels hold significant potential as electrolytes for energy storage devices. However, to fully leverage their capabilities, it is important to comprehend the atomic-level transport properties of eutectogels. This understanding will pave the way for improving their conductivity and stability, enabling widespread and efficient use in batteries. When Deep Eutectic Solvent (DES) combines with inorganic/organic matrices, eutectogels are formed, becoming advanced electrolytes. This study focuses on two key DES components: Lithium bis(trifluoromethanesulfonyl)imide as the hydrogen bond acceptor and N-methylacetamide as the hydrogen bond donor. Eutectogels are produced by combining DES with silica gel and β-poly(vinylidene fluoride-co-hexafluoropropylene) in varying ratios. Molecular dynamics analysis is used to investigate their structural properties and dynamics. The Radial Distribution Function and hydrogen bond network evolution reveal insights into the system's structural arrangement and intermolecular interactions. Transport properties are studied through the Mean Squared Displacement, providing information about species diffusion within the system. Diffusive regimes help identify the behavior of Lithium ions in the gel electrolyte, facilitating their application in

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energy storage devices. The role of ionic conductivity in the eutectogel system's performance is also explored to understand the matrix's impact on transport properties. This research aims to establish eutectogels as efficient electrolytes for energy storage applications.

1. Introduction

The design and optimization of electrolytes for efficient performance in Lithium-ion batteries (LIBs) are of utmost importance in pursuing sustainable energy systems. Batteries find various applications, such as electric vehicles [1,2], medical devices [3,4], electronics [5–8], etc. To meet the rising energy demand, batteries must be low-cost, durable, and efficient while also being environmentally benign at the end of their life [9]. LIBs have become an essential part of this system, relying on the transport of lithium ions (Li^+) via the electrolyte during electrochemical cycling, allowing the storage and release of energy. The selection of the electrolyte is crucial for determining the battery's properties, such as cycle life, energy density, capacity, and power density, which are influenced by the conductivity, viscosity, and stability of the electrolyte [10–13]. LIBs typically employ organic solvents like Ethylene Carbonate (EC), Ethyl Methyl Carbonate (EMC), Propylene Carbonate (PC), Diethyl Carbonate (DC), and many more as solvents, which have high conductivity and electrochemical stability but low boiling points, high volatility, flammability, and low thermal stability [14–16]. These drawbacks could cause hazards such as fire, toxic gas release, or thermal runaway [17,18].

To address these issues, scientists are exploring the use of Room Temperature Molten Salts (RTMS) as safe electrolytes [19,20]. RTMS is a category of electrolytes that remain in a liquid state up to 100 °C and includes Ionic Liquid (IL) and Deep Eutectic Solvents (DES). They offer several advantages over traditional electrolytes due to their low melting point [21,22], outstanding heat and chemical stability [19,23,24], reduced volatility [25], high ionic conductivity [26–28], low cost [24, 29], and safe nature [30,31]. ILs are currently being considered as a promising substitute for electrolytes [32]; however, they exhibit low conductivity, poor mixing of salts, and high viscosity compared to DES, adversely affecting energy performance [33]. Eutectogels, a class of electrolytes with unique structural and transport properties, have been proposed as promising candidates for high-performance electrolytes [34]. An Eutectogel is composed of a deep eutectic solvent (DES) and an organic/inorganic matrix [35]. DESs are a unique class of proportionate mixtures of a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD), substantially reducing the melting point compared to their original precursors [36,37]. The decrease in the melting point of DES is the result of charge delocalization during hydrogen bonding [38]. Such combinatorial characteristics lead to tunable and fascinating properties such as high thermal stability, a wide liquidus range, and the ability to form non-bonded interactions with others [39,40]. Many DESs exhibit electrochemical activity, making them a potential class of electrolytes [41–43]. They offer a variety of applications, such as being used as electrolytes in batteries [21,44,45], gas capture [46,47], biocatalysis [29], extraction and separation [48–52], medicine [53], fatty acid extraction [54], chemical synthesis [55], catalysis [56,57], as well as in electrochemistry [58,59]. The recently reported eutectogel presents a novel application of using Deep Eutectic Solvents (DES) as a gel electrolyte in lithium-ion batteries [35,60]. The combination of lithium bis (trifluoromethanesulfonyl)imide (LiTFSI) as the hydrogen bond acceptor (HBA) and N-methyl acetamide (NMAC) as the hydrogen bond donor (HBD) offers a promising DES for eutectogel [30,61]. LiTFSI is one of the most extensively utilized salts as an electrolyte in Lithium-ion Batteries (LIBs) [62]. It exhibits high solubility in polar solvents, facilitating easy dissolution in the electrolyte. With a high melting point (Tmelting = 507.15 K) [63], it remains stable at elevated temperatures without decomposing. LiTFSI contributes to lower ion transport resistance and enhances battery performance due to its low viscosity and high ion conductivity [64]. Furthermore, it is compatible with a wide range of electrode materials, ensuring its usage in tailored applications. In comparison to its counterparts, such as lithium perchlorate or lithium hexafluorophosphate, LiTFSI is less prone to generating flammable gases when overcharged, enhancing battery safety [65]. NMAC exhibits the characteristics of a green solvent, displaying physical properties similar to water, with negligible vapour pressure ($p^0 = 5 \times$ 10-4 bar at 313 K), increasing stability by minimizing the possibility of evaporation during battery operation [66]. Its high relative permittivity (ε = 178.5 at 303.15 K) demonstrates high conductivity and capacitance [67]. The interaction between Li^+ ions and the carbonyl group of NMAC weakens the relevant bonds, improving ion transport by increasing the dissociation of both positive and negative ions [68]. This interaction plays a crucial role in the development of new and high-performance electrolytes for solid-state batteries, offering improved ionic conductivity and battery performance.

Silica gel is widely used in batteries due to its large surface area [69] and significant electrolyte contact interface that enhances ionic conductivity [70]. The material's chemical and mechanical stability [71], along with its porosity, facilitates ion transfer [72], making it an ideal choice for mass production. In solid-state batteries, silica gel is often employed as a matrix to prevent direct contact between the anode and cathode, acting as a solid electrolyte separator. β-Poly(vinylidenefluoride hexafluoropropylene) (β-PVDF-HFP) is a frequently used polymer in the fabrication of solid polymer electrolytes (SPEs) for lithium-ion batteries. First used in 1996 [73], it is now the most commonly used polymer in the lithium polymer battery market [74]. Previous studies have shown that the mechanical strength, flexibility, and stability of the battery increase with the use of β-PVDF-HFP [75–79]. The incorporation of the -CF3 moiety within the hexafluoropropylene (HFP) monomers, in conjunction with vinylidenefluoride (VDF) monomers, leads to increased amorphicity and steric hindrance within the resulting molecule [80,81]. This amorphous domain, generated in an otherwise crystalline PVDF structure, enables the structure to trap large quantities of liquid electrolytes, lowers its melting temperature, and results in higher ionic conductivity [82,83]. The high dielectric constant (ε = 8.4) [84,85] and low toxicity of β-PVDF-HFP improve the electrolyte's conductivity. Compared to other structures, β-PVDF-HFP has a highly crystalline structure, improving ion transport across the electrolyte. The strong intramolecular forces resulting from its high melting point and ordered structure allow β-PVDF-HFP to act as a matrix for the DES. This advancement holds great promise for the design of electrolytes intended for room-temperature Li-ion batteries [86]. Furthermore, studies have been conducted on the physico-chemical properties of DESs formed by various lithium salts and N-methyl acetamide (NMAC) [30]. The addition of DES and the matrix in the eutectogel offers numerous benefits. Silica gel, added to the system, provides structural support and enhances its stability. Furthermore, the inclusion of PVDF-HFP contributes to the mechanical stability of the eutectogel while also imparting flexibility, allowing its safe usage even after experiencing abusive operations. These additions result in a wide electrochemical window and high thermal stability. Importantly, the non-flammable nature of the components ensures that the eutectogel is safe to handle. Developing methods to study the macroscopic properties of electrolytes is challenging without understanding molecular-level interactions. Characterizing solvation structures and designing electrolytes often relies on trial and error. Techniques like Raman spectroscopy provide limited insights into atomic-level interactions. Molecular Dynamics (MD) simulations enable accurate computations and study atomistic interactions,

bridging the gap between atomic and bulk electrolyte characteristics. Looking at the development over the past five years, it is evident that eutectogel holds the promise to be the next-generation electrolyte for energy storage devices. Providing prediction routes for computing structural and transport parameters through MD simulation is crucial due to the lack of systematic theoretical work on Li^+ based DES despite their potential applications.

MD simulations have been employed to investigate the transport properties of deep eutectic solvents (DES) based on LiTFSI for electrolyte applications. One study focused on a ternary DES system prepared with choline chloride, urea, and glycerol. The addition of a small quantity of glycerol was observed to enhance the conductivity of the DES [87]. Another set of MD simulations aimed to understand the effect of localized diffusion within a transient cage and cage-to-cage jump diffusion of acetamide-based DES. Cage-to-cage jump diffusion aids in determining transport properties, while localized diffusion within a transient cage provides insights into confinement effects in the DES [88]. Furthermore, the solubility of lithium nitrate ($LiNO_3$) is increased using the DES

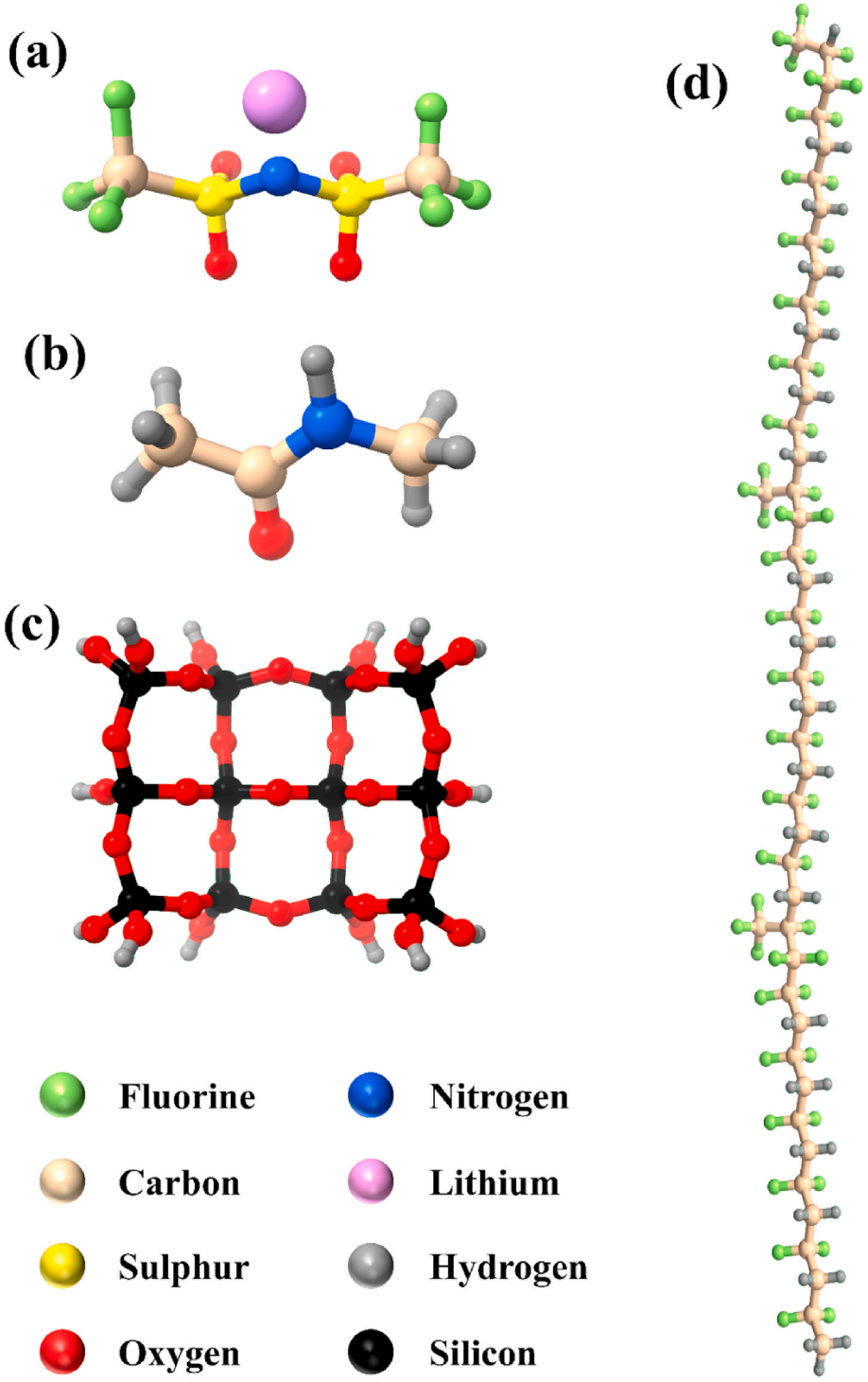


Fig. 1. Molecular structures of the components used for the simulation studies. It shows the ball and stick model of (a) LiTFSI, (b) NMAC, (c) Silica gel and, (d) β-PVDF-HFP.

electrolyte made of LiTFSI and N-methylacetamide (NMAC) in lithium metal batteries, up to 4 wt%. The strong interactions between N-methylacetamide and lithium nitrate result in a stable anion-rich solid electrolyte interphase layer, enabling uniform lithium deposition and significantly improving number of cycles compared to conventional carbonate electrolytes [89].

Despite being a state-of-the-art electrolyte, the lack of understanding of structural and transport properties and, thereby, the ion-transportation mechanism has hampered the development of eutecto-gel for commercial application. To address this knowledge gap, a comprehensive investigation of the structure and transport properties of Li-based eutectogels using all-atom MD simulation has been attempted in this work. Two different systems of eutectogels, namely LiTFSI-NMAC-based DES with a silica gel matrix and LiTFSI-NMAC-based DES with a β-PVDF-HFP matrix, are studied. Additionally, the influence of the matrix on the structural and transport properties of the eutectogel will be explored. To gain insight into the interactions and dynamics of the eutectogels at the atomic level, detailed analyses such as radial distribution function (RDF), hydrogen bond (H-bond), and mean square displacement (MSD) are performed. This research aims to enhance the understanding of the fundamental properties and behaviour of DES-based eutectogels. By elucidating the atomistic-level transport mechanisms and intermolecular interactions, the challenges associated with DES electrolytes are addressed. Ultimately, this work contributes to the development of optimized eutectogels with improved performance as electrolytes for energy storage applications. The study of two different matrices comprising organic and inorganic components will reveal valuable insights related to the design and optimization of eutectogels in the development of high-performance Li-ion batteries.

2. Computational methodology

2.1. System identification

The DES is prepared through the combination of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) and N-methyl acetamide (NMAC) in a molar ratio of 1:4. The study encompasses two categories of eutectogels, namely LiTFSI-NMAC/silica gel and LiTFSI-NMAC/β-PVDF-HFP. The structures are generated using the Avogadro package [90]. Fig. 1 shows the molecular structure of the components used for the study. The LiTFSI-NMAC/silica gel eutectogels consist of a LiTFSI-NMAC DES integrated within a silica gel matrix. The LiTFSI-NMAC/silica gel eutectogel further differentiates into five eutectogels based on the molar ratio of DES and silica gel. Similarly, the LiTFSI-NMAC/β-PVDF-HFP category also divides into five eutectogels based on the molar ratio of DES and β-PVDF-HFP. In total, ten eutectogels are analyzed to establish correlations in structural properties and dynamic behaviour. The molecules are visualised using the UCSF ChimeraX package [91,92]. Table S1 of supplementary materials provides information regarding the composition ratios of DES and matrix within the eutectogels.

2.2. System setup

The initial configuration of the system is generated using the PACKMOL package [93]. It arranges all the atoms in a box with high volume and low density. Then, it shrinks progressively till the point where it cannot generate a new box that can have all the atoms without overlapping. The partial charge of each atom is calculated using the Gasteiger method [94]. Six units of PVDF are connected to produce the β-PVDF-HFP structure, followed by the addition of an HFP unit. Later, the oligomer units are connected to create a polymeric chain. The number of initial molecules and the respective system details are provided in Table S2 of supplementary materials.

2.3. Force field

In current simulations, the Optimized Potentials for Liquid Simulations (OPLS) Force field [95,96] is used to determine the bonded and non-bonded interactions among atoms. The functional form of the force field is given by the following equation.

$$V_{total} = \sum_{bonds} K_r\left(r - r_{eq}\right)^2 + \sum_{angles} k_\theta\left(\theta - \theta_{eq}\right)^2 + \sum_{dihedrals} k_\chi\left[1 + cos\left(n\chi - \delta_\chi\right)\right] + \sum_{improper} k_\Psi[1 + cos(n\Psi - \delta_\Psi)] + \sum_{i}^{on\ a}\sum_{i}^{on\ b}\left\{\frac{q_i q_j e^2}{r_{ij}} + 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right]\right\} f_{ij} \quad (1)$$

Where, V_{total} is the total potential energy which is summation of bond stretching energy, angle bending energy, torsional energy and non-bonded energy. K_r, k_θ, k_χ & k_Ψ are bond stretching constant, bending angle energy, proper dihedral constant and improper dihedral constant, respectively. r and r_{eq} are length of bond & standard length of bond, respectively. θ and θeq are angle between the molecule & standard angle, respectively. $n\chi$ & $n\Psi$ are periodicity parameters for the torsional and improper rotations, respectively. $\delta\chi$ & $\delta\psi$ are phase shift. εij & σij are L-J potential parameters for non-bonded interactions. rij is the distance between the ith & jth atoms. q_i & q_j are charges on the ith & jth atoms. The value of f_{ij} is given by the equation (2),

$$f_{ij} = \begin{cases} 0.5, if\ i,j = 1,4 \\ 1.0, otherwise \end{cases} \quad (2)$$

2.4. MD simulation framework

The MD simulation consists of energy minimization, equilibration, which contains NVT and NPT and a final production run. All MD simulations are performed using the GROMACS package [97], and the trajectories are analyzed with the Visual Molecular Dynamics(VMD) package [98]. Energy minimization is performed to lower the potential energy of the system and help the molecules achieve a stable conformation. By reducing the potential energy, the stability of the system is enhanced. The energy minimization employs the Steepest Descent algorithm with a convergence threshold of 1000 kJ/mol nm. It is currently being executed for 5×10^6 steps, with a step size of 10^{-3} nm. A leap-frog integrator [99] is used to integrate all the equations of motion with a time step of 2 fs in the equilibration and final production run. Hydrogen bonds are constrained by the linear constraint solver (Lincs) algorithm [100] in the equilibration stage. The long-range electrostatic interactions are treated using the particle-mesh Ewald (PME) method [101]. Van der Waals interactions are cut off at a distance of 1.2 nm. To update the neighbour grid list, the verlet algorithm is used with a frequency of 20 steps. Equilibration is a two-stage process, the first stage is a canonical ensemble (NVT), and the second stage is an isothermal isobaric ensemble (NPT). During the equilibration and final production stage, temperature is maintained using the velocity-rescaling thermostat (V–rescale) [102]. Also, the pressure is maintained in NPT and the final production stage using the Parrinello-Rahman barostat [103]. NVT ensemble is performed for 1 ns at 300 K temperature with a time step of 2 fs. During the NVT equilibration process, the annealing is carried out by increasing the temperature from 300 to 370 K with a linear increase over time at a rate of 0.1 K ps^{-1}. NPT ensemble is performed for 19 ns at 300 K temperature and 1.013 bar pressure with the time step of 2 fs. The annealing is done by decreasing the temperature from 370 to 300 K with a linear increase over time at a rate of 0.01 K ps^{-1} in the NPT equilibration. During the NPT equilibration process, the total energy data, density of the system and volume of the simulation box with respect to the time are given in Figs. S1, S2, and S3, respectively, of the

supplementary materials. The NPT equilibration result shows that the energy, density, and volume of the system are nearly constant. The NPT result concludes that the system is well equilibrated and can be moved to the production run. The final production run consists of a 15 ns production run with a 300 K and 1.013 bar temperature and pressure, respectively. The system's coordinates and velocities are recorded every ten ps for later examination. Fig. S4 of supplementary materials represents the final snapshot of eutectogels after 15 ns, whereas Figs. S5 and S6 represent the time-resolved geometries of the systems, respectively.

2.5. Analysis of trajectories

2.5.1. Radial distribution function (RDF)

The radial distribution function (RDF) estimates the structural distribution of atoms in a system around a reference atom. It describes the frequency of locating an atom in a material or system at a fixed distance from the reference atom. RDF is used to study the arrangement and interactions between the ions with the electrolytes and electrode surface. RDF is calculated by equation (3),

$$g_{AB}(r) = \frac{\langle \rho_B(r) \rangle}{\langle \rho_B \rangle_{local}} = \frac{1}{\langle \rho_B \rangle_{local}} \frac{1}{N_A} \sum_{i \in A}^{N_A} \sum_{j \in B}^{N_B} \frac{\delta(r_{ij} - r)}{4\pi r^2} \tag{3}$$

Where, $g_{AB}(r)$ is RDF, $\langle \rho_B(r) \rangle$ is the average density of atoms of B which are at a distance r from atoms of A, $\langle \rho_B \rangle_{local}$ is the local average density of atoms of B which are part of a sphere of radius dr from the atoms of A, N_A & N_B is the total number of atoms of A and B respectively, r is the distance between the central atom and other atoms, r_{ij} is the distance between ith and jth atoms & $\delta(r_{ij} - r)$ is the Dirac delta function. It is defined as

$$\delta(r_{ij} - r) = \begin{cases} 1, (r_{ij} - r) = 0 \\ 0, (r_{ij} - r) \neq 0 \end{cases} \tag{4}$$

2.5.2. Coordination number (CN)

CN is defined as the number of atoms of B around the reference atom A. It gives more knowledge on the local structure and bonding conditions. For instance, a high coordination number might signify tightly packed atoms or molecules and strong bonds, whereas a low coordination number might signify a more porous structure and weaker bonds. *CN* is represented by equation (5)

$$CN(r) = 4\pi\rho N \int_0^r g(r) r^2 dr \tag{5}$$

Where, ρ is the number density of particles, N is the quantity of atoms in the system and $g(r)$ is the RDF. *CN* can be calculated by integrating the RDF over a spherical shell of radius r centered on atom A, and dividing the result by the volume of the simulation cell.

2.5.3. Hydrogen bond (H-bond)

Strong hydrogen bond (H-bond) is formed between Hydrogen and high electronegative atoms namely Fluorine (F), Oxygen (O), Nitrogen (N) only in the given system. To form H-bond the distance between both the atoms should not exceed 3.5 Å and the angle should not exceed 30°. The number of hydrogen bonds is calculated using equation (6),

$$C_{HB} = \frac{\langle H(\tau) H(\tau + t) \rangle}{\langle H(\tau) H(\tau) \rangle} \tag{6}$$

Where, C_{HB} is the number of hydrogen bonds, $H(\tau)$ and $H(\tau+t)$ are the H-bond order parameter at time τ and $(\tau+t)$ respectively. In the simulations of LiBs, hydrogen bonding can be quite important. The organic electrolyte solution used in LiB often includes atoms that may form H-bonds with other species present in the electrolyte. H-bonding affects the solvation structure and dynamics of the ions, as well as the stability of the electrolyte. In a LiB simulation, the RDF of the species involved may be used to calculate the strength and rate of H-bonds. The likelihood of discovering a molecule or ion at a specific distance from a reference molecule or ion is quantified by the RDF. When the RDF exhibits a peak at a certain distance, there is a significant likelihood of discovering the two species which may be the result of H-bonding. Increased solvation energy and a more stable electrolyte can result from strong H-bonds among the solvent molecules and the Li-ions, which can boost the battery performance. However excessive H-bonding can also result in aggregate formation, which can obstruct ion transport and lower battery performance. To maximise battery performance, it is crucial to balance the strength of H-bonding in the electrolyte.

2.5.4. Mean square displacement

Mean Square Displacement (MSD) is the average of squared displacement of atom or molecule of from its original position over a period of time. It is used to calculate the diffusivity of the lithium ions in the electrolyte. The lithium-ion migration has to be the utmost importance in lithium-ion battery system. The MSD is calculated for lithium ions and which are strongly interacting with lithium ion. It provides information about the movement of moieties across the electrolyte. Diffusion coefficient (D_A) is computed using the Einstein relation [104] given by the equation (7):

$$\lim_{t \to \infty} \langle \left\| r_i(t) - r_i(0) \right\|^2 \rangle_{i \in A} = 6 D_A t \tag{7}$$

Where $r_i(t)$ and $r_i(0)$ are the location of centre of atom at time t and 0, respectively. Also, the parameter δ provides insight about the atom or molecule behaviour and can be calculated using equation (8) [105],

$$\delta = \lim_{t \to \infty} \frac{d[log(\Delta r^2)]}{d[log(t)]} \tag{8}$$

If the value of δ is equal to 1, it signifies a diffusive behavior of atoms. In this case, the atoms exhibit random and uncorrelated motion. When δ is less than 1, it indicates sub-diffusive behavior. This implies that the atoms experience hindered or restricted movement, possibly due to various factors such as obstacles or intermolecular interactions. On the other hand, if δ is greater than 1, the atoms demonstrate super-diffusive behavior. This suggests enhanced or accelerated movement compared to ordinary diffusion. The atoms may undergo more extensive displacements or exhibit correlations in their motion, leading to phenomena like anomalous transport.

2.5.5. Ionic conductivity

Ionic conductivity is defined as a measure of the ability of ions to move and transport charge through the system. It is calculated using electric-current autocorrelation function (ECACF) using Green-Kubo method given by equation (9) [106],

$$\sigma_{GK} = \frac{1}{3 K_B V T} \int_0^\infty \langle \vec{j}(t) . \vec{j}(0) \rangle dt \tag{9}$$

where, σ_{GK} is the ionic conductivity calculated using the Green-Kubo method, K_B, V and T are Boltzmann constant, Volume of final simulation box and Temperature of the system. $\vec{j}(t)$ is the current autocorrelation function at time t.

3. Results and discussion

3.1. Structural properties: radial distribution function and coordination number

3.1.1. Silica gel based eutectogels

Fig. 2 shows a series of RDFs among various moieties of HBA and HBD of DES. Similarly, Fig. 2 a. shows the RDF curve between lithium ion and the nitrogen present in TFSI. The first solvation shell is observed at 2.65 Å with a *CN* number of 0.41 for ETG – 1 to ETG – 4 and 0.2 for

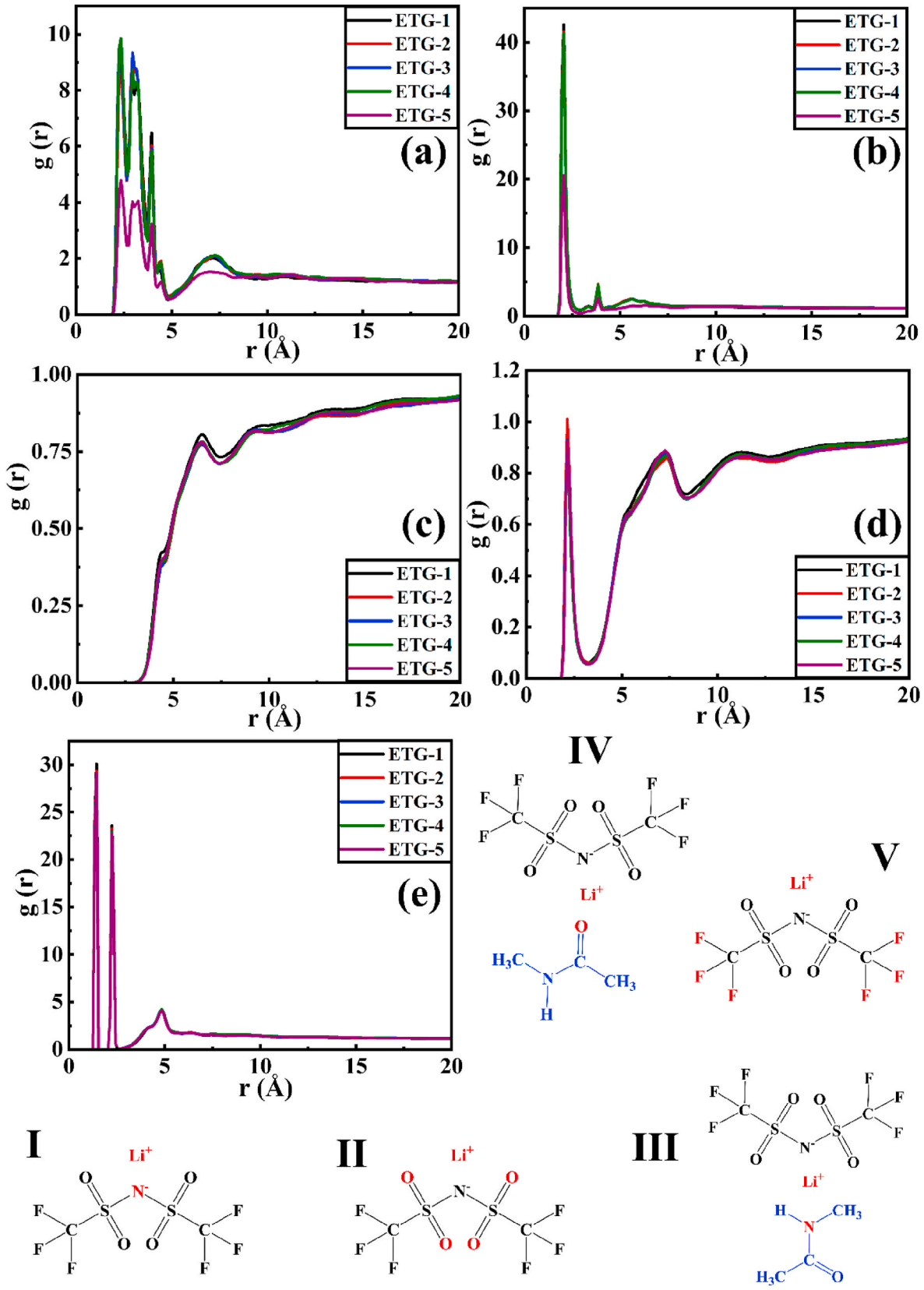


Fig. 2. RDF of silica gel based eutectogel system (a) Li–N (LiTFSI), (b) Li–O (LiTFSI), (c) Li–N (NMAC), (d) Li–O (NMAC), and (e) Li–F (LiTFSI). I. schematic representation of Li–N (LiTFSI) interaction, II. schematic representation of Li–O (LiTFSI) interaction, III. schematic representation of Li–N (NMAC) interaction, IV. schematic representation of Li–O (NMAC) interaction, and V. schematic representation of Li–F (LiTFSI) interaction.

ETG - 5. It is also observed that by the ETG – 1 to ETG – 4, the DES concentration in the solution increases, and the RDF value remains the same whereas in the ETG – 5, with the increasing DES concentration, the RDF value decreases significantly. The second solvation shell is observed at 3.65 Å with a *CN* number of 1.45 for ETG – 1 to ETG – 4 and 0.73 for ETG - 5. The second solvation shell also shows similar behaviour to the first solvation shell. The width of the second solvation shell in the ETG – 1 to ETG – 4 shows similar trend, whereas the width of the ETG – 5 is marginally smaller in comparison to other ETGs. It signifies that the ETG – 5 shows more compact behaviour than other ETGs. Fig. 2 b. shows the RDF curve between the Lithium ions and oxygen atoms present in the TFSI molecules. The first solvation shell is observed at 2.85 Å with a *CN* number of 3.37 for ETG – 1 to ETG – 4 and 1.69 for ETG - 5. The ETG – 5 shows a lower solvation shell than other ETGs. It signifies that DES molecules are less able to form strong interactions with matrix. The second solvation shell is observed at 4.15 Å with a *CN* number of 1.91 for ETG – 1 to ETG – 4 and 0.99 for ETG - 5. Fig. 2 c. shows the interaction between lithium-ions and nitrogen atoms in NMAC. The first solvation shell observes at 7.45 Å with a *CN* number of 5.13, and its RDF value is less than one. After the first solvation shell, the RDF value increases slowly, which signifies the formation of a small size cluster. It suggests the non-bonded interaction between Li and N atoms. Fig. 2 d. shows the RDF between lithium-ions and oxygen atoms in NMAC. The first solvation shell occurs at 3.25 Å. The second solvation shell is observed at 8.35 Å. The RDF exhibits values less than unity for both the first and second solvation shells, indicating the formation of small-sized clusters. This observation is further supported by the *CN* analysis. The *CN* of the first solvation shell ranges around 0.15 for ETG - 1 to ETG - 5, while the *CN* of the second solvation shell is approximately 8.35. The high *CN* value suggests strong interactions between Li^+ and O atoms within the second solvation shell. Fig. 2 e. shows the RDF value between the lithium ions and fluorine atoms present in the TFSI. It shows the two consecutive solvation shells under the 2.55 Å with a *CN* number of 1 and 3. This consecutive solvation occurs because the fluorine atoms are present in the TFSI anion on both tail sides. Fig. 2 signifies that the solvation occurs because of electrostatic interaction between the lithium ions and other moieties (shown in red). It is also observed that the lithium-ion oxygen moiety of TFSI ions strongly interacts with each other. This is because the oxygen atoms of TFSI have the maximum negative charge of other DES atoms.

Figs. S7 and S8 of supplementary material show the RDF curve between different moieties of silica gel and different moieties of DES. The solvation shells occur because of the weak, non-bonded interaction between the molecules. The non-bonded interaction can be attractive and repulsive interaction based on the charge of the molecules. The RDF value of each curve is less than unity or marginally greater than unity. Fig. S7 b., S7.d., and S8 show the sharp solvation shells between oxygen moiety present in different DES molecules to silica moiety and oxygen moiety of silica gel to lithium-ion of LITFSI. High *CN* is observed in the second shell of both systems. It suggests that the solvent molecules are arranged in a regular pattern, forming stable and ordered structures around the matrix and strong non-bonded interactions. Fig. S7 e. shows the RDF curve between silicon atoms in silica gel and fluorine atoms present in LiTFSI. The average coordination number observed for the first and second shell is approximately 2 and 9.15, respectively. These findings indicate the formation of an ordered structure around the matrix. Fig. S7 f. shows the RDF curve between lithium-ions in LiTFSI and oxygen atoms present in silica gel. The estimated *CN* is very low, and it decreases from ETG – 1 to ETG – 5, indicating a decrease in the matrix concentration. The solvation shells come because of the attractive non-bonded interactions between the molecules.

From Fig. 2, Fig. S7, and S8, the lithium-ion of the LiTFSI strongly interacted with different moieties of TFSI and oxygen of silica gel. The oxygen atom of silica gel plays a crucial role in the mobility of lithium ions. Among the DES moieties, the oxygen moiety of TFSI shows the strongest interaction with the lithium ions. It is observed that ETG – 1 to ETG – 4 show similar RDF values between the lithium-ion and oxygen moiety of the TFSI, whereas ETG – 5 shows different RDF values. In the DES interactions, the ETG – 5 shows less interaction than other ETGs. In the organic gel matrix interactions, the ETG – 5 shows higher interactions than other ETGs. It signifies that the ETG – 5 is more interacting with the silica gel matrix than the other DES molecules. Table S3 gives the RDF and *CN* values at different coordination shell of different eutectogels of silica gel based eutectogel.

3.1.2. *β-PVDF-HFP based eutectogel*

Fig. 3 a. shows the RDF curve between lithium atoms in TFSI and nitrogen atoms in NMAC atoms. The first solvation shell occurs due to the strong electrostatic interaction between the atoms. The RDF values increase with a decrease in the concentration of the DES in the system. The *CN* values are nearly constant. It is forming a more stable and ordered structure. Fig. 3 b. shows the RDF between lithium atoms and oxygen atoms in NMAC atoms. The first solvation shell is observed around 3.25 Å. The RDF values are less than unity, suggesting the formation of clusters. The estimated *CN* are 0.37, 0.44, 0.47, 0.35, and 0.49 for ETG – 1 to ETG – 5, respectively. This is because of the strong electrostatic interaction between the molecules. Fig. 3 c. shows the RDF between lithium atoms and nitrogen atoms in TFSI. The first and second solvation shell occurs at 2.55 Å and 3.65 Å, respectively. The RDF value increases for both shells from ETG – 1 to ETG – 5. The *CN*s for the first shell are 0.19, 0.14, 0.10, 0.09, and 0.13, while for the second shell, they are 1.32, 1.25, 1.26, 1.22, and 1.21, respectively. Fig. 3 d. shows the RDF between lithium atoms and oxygen atoms in TFSI. The first solvation shell occurs at 2.95 Å. A decrease in *CN* is observed in the first shell. The second solvation is observed at 4.25 Å. Similar to the first shell, a decrease in CN is observed in the second solvation shell. Fig. 3. (b – d) shows a narrow width, which signifies the matrix is tightly surrounded by DES molecules that are organized in a well-defined and compact arrangement. Fig. 3 d. shows the highest RDF value, which signifies the strong electrostatic interaction, and the oxygen atom of the TFSI has the highest negative charge than other molecules.

Fig. 4 illustrates the RDF value depicting the interaction between lithium ions and fluorine moieties present in different structures. Fig. 4 a. represents the RDF curve between the fluorine moiety of TFSI to the lithium moiety of the LiTFSI. The RDF curve reveals the presence of two successive solvation shells occurring within a distance of 2.55 Å. This observation can be attributed to the strong electrostatic interaction between lithium ions and the fluorine atoms. The *CN* values for first and second shell are 1 and 3, respectively. The consecutive solvation phenomenon arises because the TFSI molecules contain fluorine atoms on both sides of their molecular structure. The lithium ions preferentially attract and interact with one side of the fluorine atoms contained within the TFSI anion, and subsequently, they engage with the other side. As a consequence, this sequential interaction leads to the emergence of discernible solvation shells. The narrow width observed in the RDF curve indicates that the solute is encircled by solvent molecules in a tightly packed and well-structured manner. Fig. 4. (b – d) shows the RDF curve between the lithium-ions and fluorine atoms present in the *β*-PVDF-HFP polymer chain. The *β*-PVDF-HFP chain has three types of fluorine present. Fig. 4 b. and 4.d signify the long-range electrostatic interactions between the fluorine atoms and lithium atoms. Fig. 4 c. shows the first solvation shells at 2.75 Å. The RDF value is less than 1, which signifies that the probability of finding lithium-ions at a particular distance from the fluorine atoms is lower than the average density. It is observed from Fig. 4 that lithium-ion strongly interacts with fluorine atoms in TFSI, and it also minorly interacts with fluorine atoms present in *β*-PVDF-HFP.

Fig. S9 shows the RDF between HBD and HBA in *β*-PVDF-HFP based eutectogel. Fig. S9 a. and S9.b. show the RDF curve between hydrogen atoms of NMAC, nitrogen and oxygen atoms in TFSI. The first solvation shell is observed due to the hydrogen bonding at 3.45 Å for F atoms while at 3.35 Å for O atoms. The *CN* value is approximately 0.01 for H

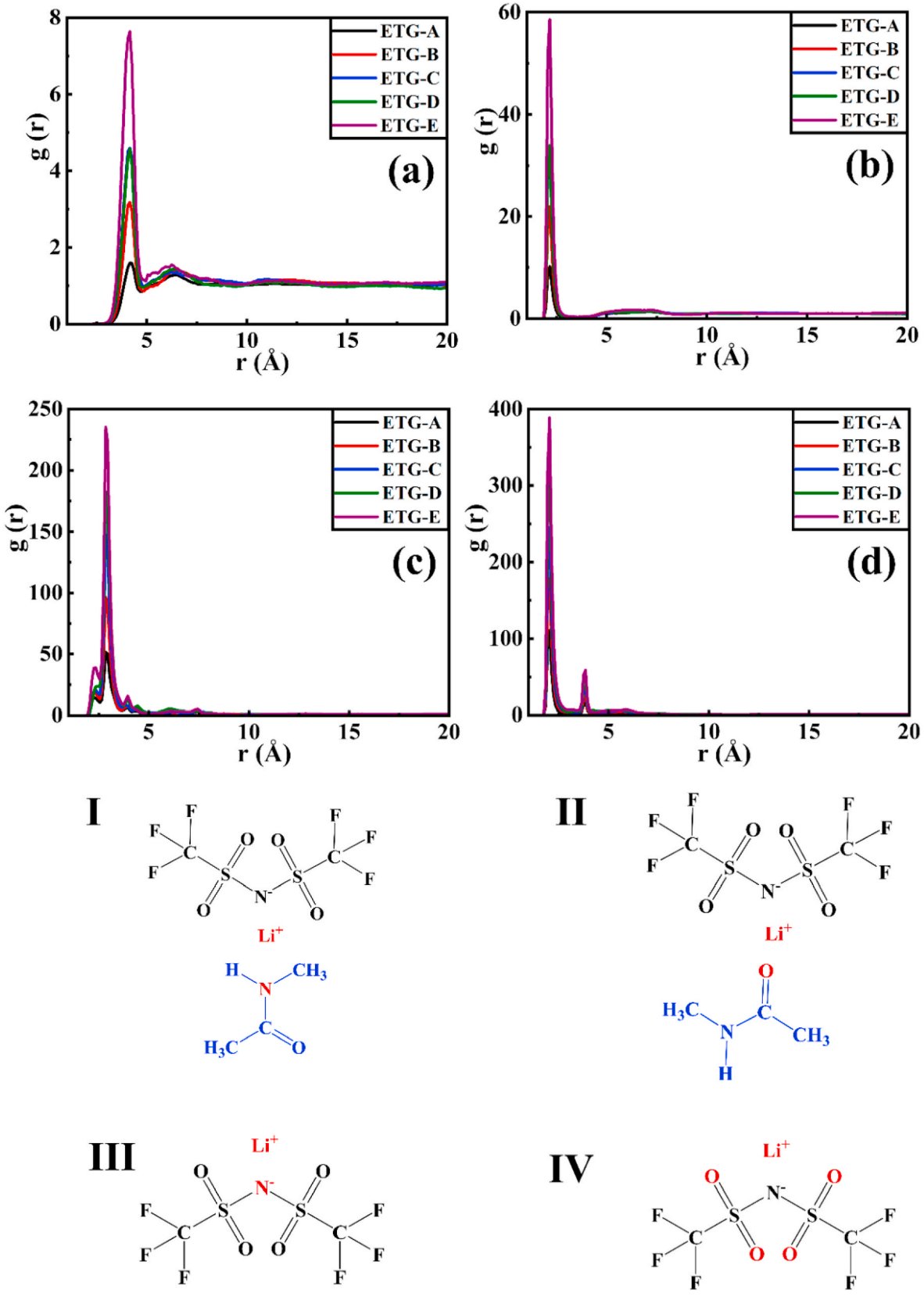


Fig. 3. RDF of *β*-PVDF-HFP based eutectogel system (a) Li–N (NMAC), (b) Li–O (NMAC), (c) Li–N (LiTFSI), and (d) Li–O (LiTFSI). I. schematic representation of Li–N (NMAC) interaction, II. schematic representation of Li–O (NMAC) interaction, III. schematic representation of Li–N (LiTFSI) interaction, and IV. schematic representation of Li–O (LiTFSI) interaction.

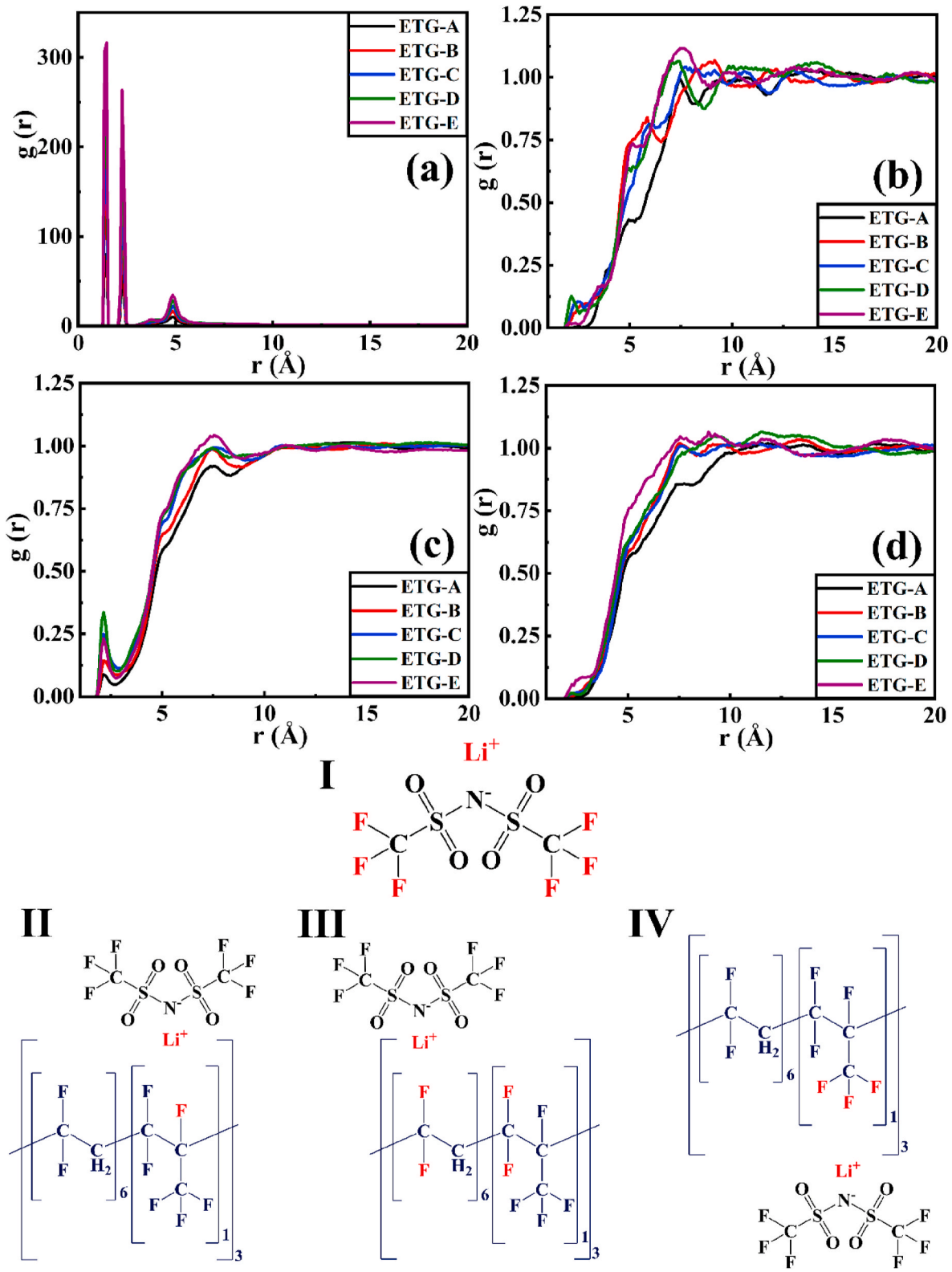


Fig. 4. RDF of β-PVDF-HFP based eutectogel system (a) Li–F (LiTFSI), (b) Li – F1 (β-PVDF-HFP), (c) Li – F2 (β-PVDF-HFP), and (d) Li – F3 (β-PVDF-HFP). I. schematic representation of Li–F (LiTFSI) interaction, II. schematic representation of Li – F1 (β-PVDF-HFP) interaction, III. schematic representation of Li – F2 (β-PVDF-HFP) interaction, and IV. schematic representation of Li – F3 (β-PVDF-HFP) interaction.

and N atoms. The *CN* decreases for ETG - A to ETG - E from 0.13, 0.07, 0.06, 0.03 and 0.04, respectively. The second solvation shell is due to the weak non-bonded interactions between the F and O atoms at 9.25 Å and 6.85 Å, respectively. The *CN* decreases for ETG - A to ETG - E from 1.43, 0.92, 0.70, 0.53 and 0.51, respectively of H and N atoms. A similar decrease in CN is observed for H and O atoms in the second shell from 2.09 to 0.83 for ETG - A to ETG – E. The RDF value of ETG – E is higher than that of other ETGs. It is creating a structure that is more stable and

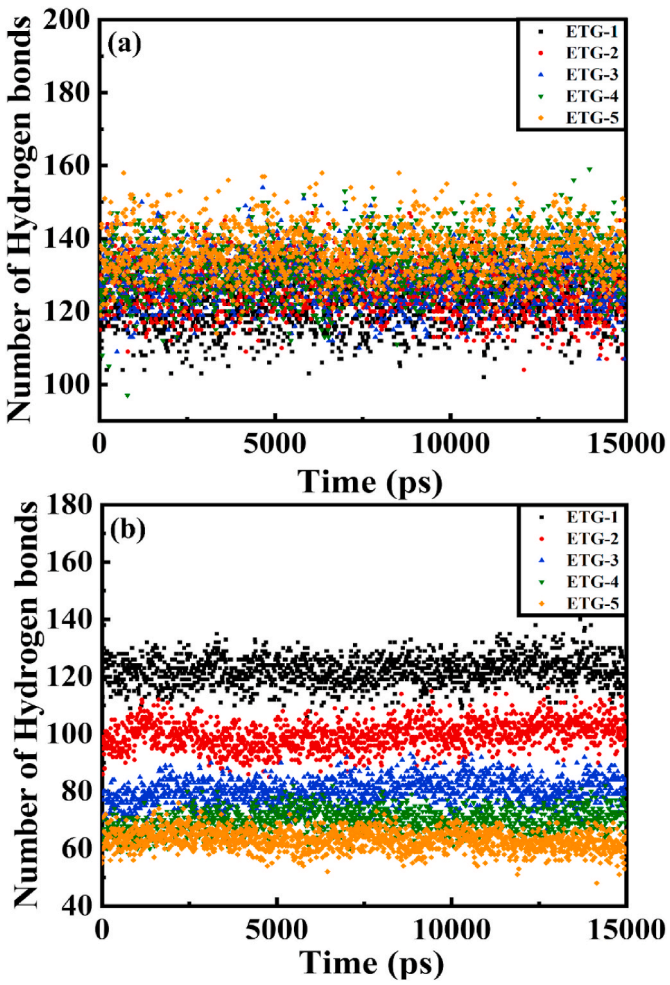


Fig. 5. Number of Hydrogen bonds in the silica gel based eutectogel (a) number of hydrogen bonds between LiTFSI and NMAC, and (b) number of hydrogen bonds between LiTFSI and Silica gel.

organized. Fig. S8 c. shows the RDF between the hydrogen atoms of NMAC and fluorine atoms of the TFSI. The first solvation shell observes at 6.75 Å. The *CN* exhibits a decreasing trend from ETG - A to ETG - E. The current values are 3.13, 2.07, 1.67, 1.21, and 1.28, indicating a gradual decline in the number of surrounding atoms in each system. It signifies that the fluorine atoms do not form the hydrogen bonding. It can only interact with weak, non-bonded interactions. Fig. S8 (d – f) shows the RDF curve between hydrogen atoms in the NMAC and three types of fluorine atoms present in the β-PVDF-HFP. It shows RDF value increases slowly. It further suggests that the hydrogen bond is not possible here. The weak, non-bonded interaction is only possible.

3.2. Hydrogen bond

Hydrogen bonds (H-bonds) play a crucial role in maintaining the structural integrity and behavior of electrolytes. The interaction between the Li^+ ion and the carbonyl group present in NMAC leads to the weakening of intramolecular H-bond in the NMAC. As a result, DES is formed [68]. Furthermore, the H-bonding interactions between DES and the matrix can influence the adhesion and wetting of the electrolyte on the electrode surface. In the battery system, the presence of a high number of H-bonds leads to an ordered structure due to the increase in intermolecular interactions. However, the rigidity impedes the movement of the ions, thereby reducing conductivity. Few H-bonds can lead to an increase in ionic conductivity but have a strong impact on battery stability. Therefore, it is important to balance the number of hydrogen bonds formed to maintain the stability of the structure and ionic conductivity throughout the system.

Fig. 5 shows the number of H-bonds formed in the silica gel based eutectogel. NMAC acts as both an H-bond donor (C═O) and an H-bond acceptor (-NH) [107]. Fig. 5 a. illustrates the fact that the number of H-bonds is constant between NMAC – LiTFSI as they are proportional to the DES formation molar ratio, and it is constant. The average H-bond formed between NMAC – LiTFSI for ETG – 1 to ETG – 5 are 122, 126, 128, 133, and 136, respectively, given in Table 1. Here, the NMAC acts as a hydrogen bond donor, whereas TFSI acts as a hydrogen bond acceptor. The average number of H-bonds increases because the concentration of DES increases from ETG-1 to ETG-5. Fig. 5 b. shows the number of H-bonds between the LITFSI and silica gel matrix. The silica gel matrix works as a H-bond donor, and the LITFSI acts as a H-bond acceptor. Analyzing the Fig. 5 b. it is evident that, at 11.5 ns, an increment in the H-bond is observed in ETG – 1 between LiTFSI and silica gel. Similarly, decreases in H-bonds are observed after 8.9 ns and 6.1 ns in ETG – 3 and ETG – 4, respectively, between LiTFSI and silica gel. Fig. 5 b. shows a decrease in the number of H-bonds between LiTFSI – silica gel as the molar ratio of silica gel is reduced. The H-bond is mainly due to the presence of hydroxyl moiety present on the surface of silica gel. The average number of H-bonds for ETG – 1 to ETG – 5 are 122, 100, 80, 70, and 62, respectively, given in Table 1.

Similarly, in β-PVDF-HFP based eutectogel, even though highly electronegative fluorine atom is present in the β-PVDF-HFP, it is observed from RDF plot in Fig. S8 that the distance between H and F atoms is more than the threshold distance mentioned in the section 3.2 and hence no H-bond is formed between NMAC and β-PVDF-HFP. Fig. 6 a. shows the average number of H-bond between NMAC – LiTFSI for ETG – A to ETG – E are 29, 9, 5, 3 & 3, respectively, given in Table 1. Similar to silica gel based eutectogel, the dominant hydrogen bonding involved the O atom of LiTFSI and the H atoms of NMAC, as evident from Fig. 6 b. The average number of H-bond between NMAC – LiTFSI for ETG – A to ETG – E are 34, 12, 6, 4, and 2, respectively, given in Table 1. Overall, the H-bonding analysis highlights the presence of significant interactions between LiTFSI, NMAC and the matrices in both Eutectogel systems. These interactions play a crucial role in determining the structure and dynamic properties of the eutectogel.

Table 1
The average number of H-bonds present in silica gel based eutectogel.

System	silica gel based eutectogel		β-PVDF-HFP based eutectogel	
	Eutectogel	Average number of Hydrogen Bond	Eutectogel	Average number of Hydrogen Bond
NMAC – LiTFSI	ETG – 1	122	ETG – A	29
	ETG – 2	126	ETG – B	9
	ETG – 3	128	ETG – C	5
	ETG – 4	133	ETG – D	3
	ETG – 5	136	ETG – E	3
LiTFSI – silica gel	ETG – 1	122		
	ETG – 2	100		
	ETG – 3	80		
	ETG – 4	70		
	ETG – 5	62		
NMAC – silica gel	ETG – 1	121		
	ETG – 2	86		
	ETG – 3	64		
	ETG – 4	53		
	ETG – 5	45		
NMAC – NMAC	ETG – 1	300	ETG – A	34
	ETG – 2	306	ETG – B	12
	ETG – 3	315	ETG – C	6
	ETG – 4	316	ETG – D	4
	ETG – 5	317	ETG – E	2
silica gel – silica gel	ETG – 1	39		
	ETG – 2	30		
	ETG – 3	22		
	ETG – 4	18		
	ETG – 5	17		

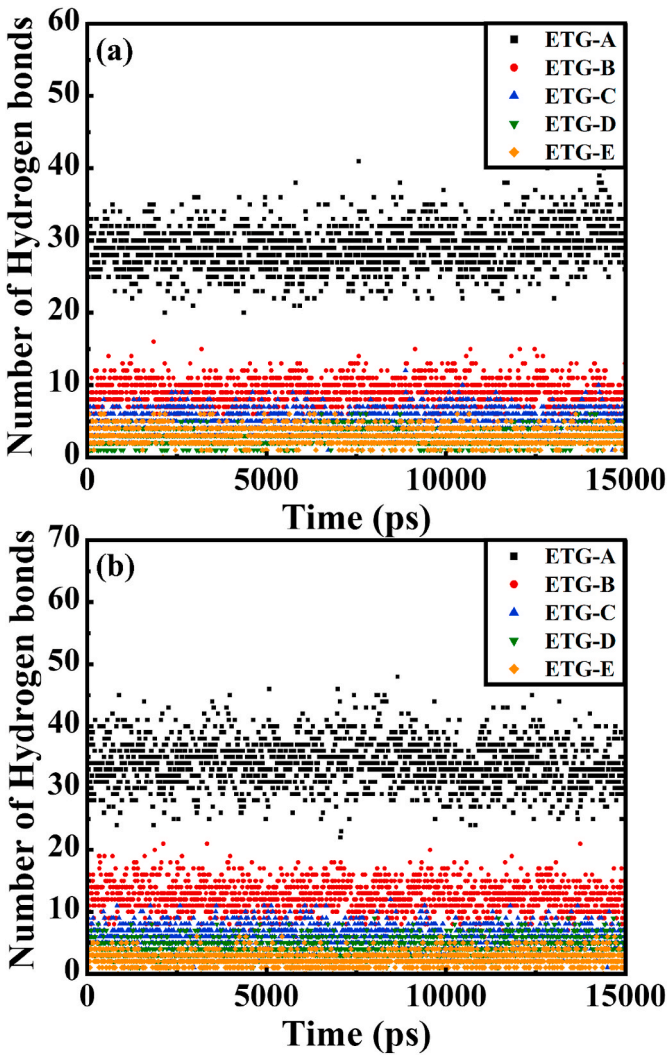


Fig. 6. Number of Hydrogen bonds in the β-PVDF-HFP based eutectogel. (a) number of hydrogen bonds between LiTFSI and NMAC, and (b) number of hydrogen bonds between NMAC and NMAC.

3.3. Dynamic behaviour: mean square displacement

Fig. 7 shows the MSD profile of the lithium atom in LiTFSI of silica-based eutectogels. Fig. 7 a. represents the MSD curve of the lithium ion in LiTFSI in ETG – 1. The MSD curve exhibits the subdiffusive behavior of the lithium ion. However, the curve can be divided into three parts. In the first part, lasting until 2.9 ns, a slow movement of the lithium ion is observed, which can be attributed to the confinement of the atoms. In the subsequent part, the δ value increases from 0.26 to 0.55 until 11.5 ns, indicating a transition where the lithium ions migrate from a more confined to a less confined environment. However, after 11.5 ns, the δ value of the MSD curve decreases to 0.31. Simultaneously, an increase in H-bond formation between LiTFSI and silica gel is observed, which enhances the rigidity of the eutectogel. This rigidity hinders the movement of the lithium ions. Fig. 7 b. illustrates the MSD curve as a function of time for the lithium ion in LiTFSI of ETG – 2. The MSD curve exhibits four distinct regions. In the first region, lasting until 7.5 ns, the MSD curve shows slow movement, indicating sub-diffusive behavior of the Lithium ion. This behavior is attributed to the random motion and interactions with the surrounding molecules. Following this, the MSD curve gradually increases until 12.3 ns, indicating super-diffusive behavior. This occurs when the atoms are confined within a restricted or crowded environment or when the solvent has a non-uniform structure causing correlated particle movement. During the time interval between 12.3 ns and 13.8 ns, the MSD curve decreases, suggesting subdiffusive behavior due to hindered diffusion caused by collisions with other particles. Finally, the curve displays a linear increase in MSD value with time, representing the diffusive behavior of the Lithium ion. It represents the diffusive behaviour of lithium-ion. Here, the self-diffusivity value is 1.2×10^{-12} m^2 s^{-1}. The lithium ion is currently attempting to reach the diffusive region, leading to a transformation from subdiffusive to superdiffusive behavior. Subsequently, it returns to the subdiffusive motion before eventually adopting a diffusive behavior. Fig. 7 c. illustrates the MSD curve depicting the motion of lithium ions in LiTFSI of ETG – 3 over time. The MSD curve exhibits a gradual motion of the lithium ions with subdiffusive behavior. After 8.9 ns, there is an increase in the δ value from 0.31 to 0.38. Simultaneously, a decrease in the number of H-bonds between LiTFSI and silica gel is observed, leading to a less confined structure. This marginal decrease in confinement facilitates an increase in the movement of the lithium ions. Fig. 7 d. illustrates the MSD curve representing the motion of lithium ions in LiTFSI of ETG – 4 over time. Similar to ETG – 3, the lithium ions exhibit a gradual movement throughout the system until 6.1 ns, with a δ value of 0.37, indicating subdiffusive behavior. Subsequently, there is a sudden increase in the δ value to 0.79. This abrupt increase is attributed to a sudden decrease in the number of hydrogen bonds between LiTFSI and silica gel. As a result, the confinement in the system is reduced, allowing a less compact place for lithium ions to move. Fig. 7 e. illustrates the MSD curve representing the motion of the lithium ion in LiTFSI of ETG – 5 over time. In the initial region the system exhibits slow movement, indicating subdiffusive behavior until 11.2 ns. This behavior is a result of the confinement of the lithium ion within its surroundings. However, the lithium ion experiences a sudden increase and enters the superdiffusive region. This is because the interaction force between the oxygen moiety of silica and the lithium-ion enhances, as shown in Fig. S8 of the supplementary material. As a result, it neglects the interaction force between the lithium-ion and the different moiety of TFSI ion. As progress is made from ETG-1 to ETG-5, the MSD value is gradually increased, indicating a higher degree of lithium-ion movement and diffusion. This is because the mean volume of each system is nearly equal. The concentration of TFSI ions increases and the TFSI ion start to repel each other. As a result, the electrolyte doesn't form clusters, and the mobility also increases. This trend is consistent with the expectation that higher concentrations of lithium ions would lead to increased mobility. However, with ETG – 3, a sudden and unexpected drop in the MSD value is observed. This unexpected behavior suggests that the movement of lithium ions within the eutectogel is nonlinear with respect to its concentration.

Fig. S10 shows the MSD profile of the TFSI ion in LiTFSI of silica-based eutectogels. Fig. S10 a. represents the MSD curve of the TFSI ion in LiTFSI in ETG – 1. The MSD curve exhibits subdiffusive behavior of the TFSI ion. However, the curve can be divided into three parts. In the first part, lasting until 1.4 ns, there is slow movement observed for the TFSI ion, which is attributed to the confinement of the atoms. In the subsequent part, the δ value increases from 0.25 to 0.41 until 11.5 ns, indicating a transition where the TFSI ions migrate from a more confined to a less confined environment. However, after 11.5 ns, the δ value of the MSD curve decreases to 0.28. Simultaneously, an increase in H-bond formation between LiTFSI and silica gel is observed, enhancing the rigidity of the eutectogel, thereby hindering the movement of the TFSI ions. Fig. S10 b. illustrates the MSD curve as a function of time for the TFSI ion in LiTFSI of ETG – 2. The MSD curve exhibits four distinct regions. In the first region, lasting until 6.9 ns, the MSD curve shows slow movement, indicating the sub-diffusive behavior of the TFSI ion. This behavior is attributed to the random motion and interactions with the surrounding molecules. Following this, the MSD curve gradually increases until 12.2 ns, indicating diffusive behavior. The δ value of the MSD curve increases to 1. Here, the self-diffusivity value is 1.3×10^{-12} m^2 s^{-1}. This occurs when the atoms are confined within a restricted or

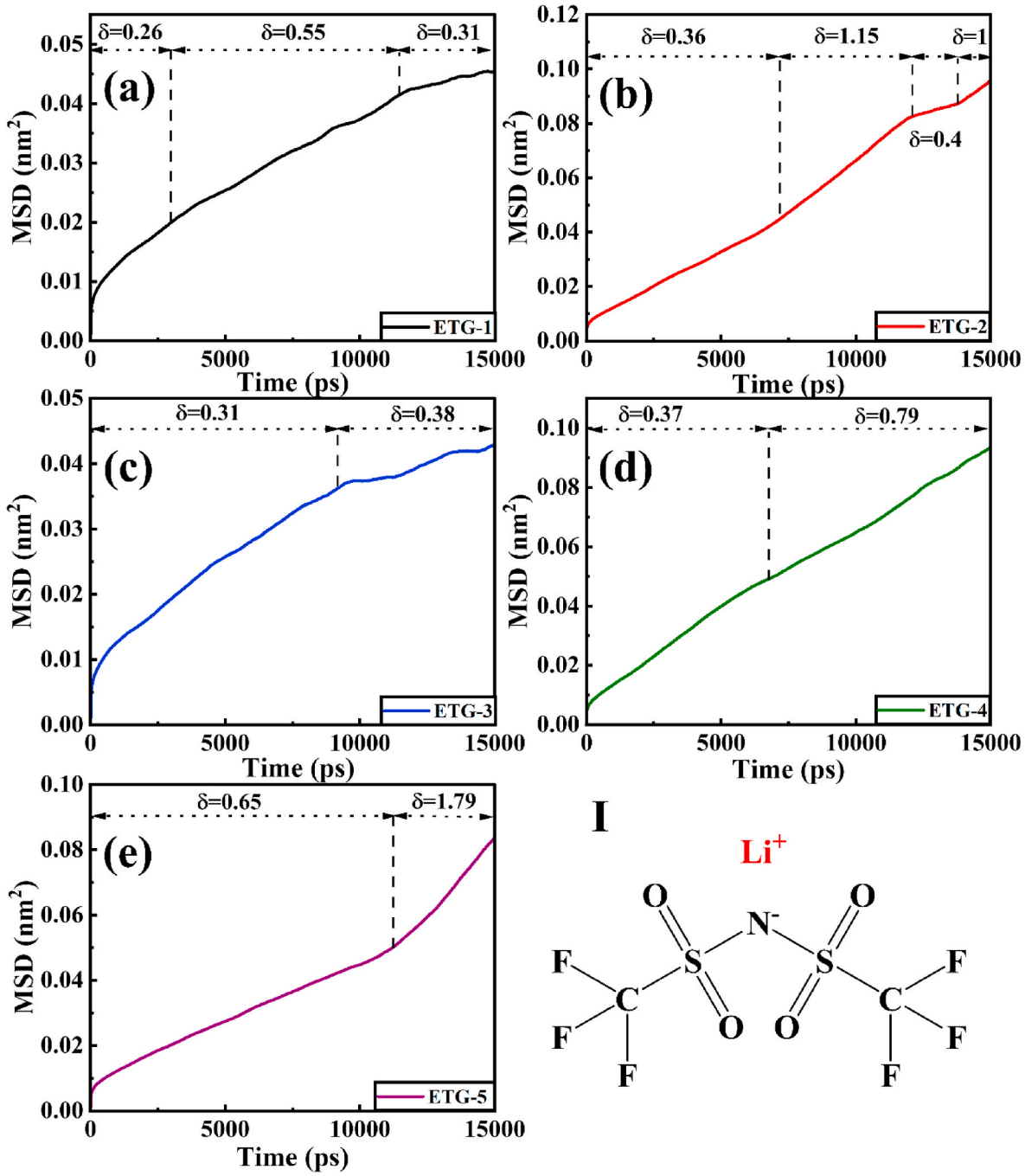


Fig. 7. The MSD curve of Li in LiTFSI in silica gel based eutectogel. (a) ETG – 1, (b) ETG – 2, (c) ETG – 3, (d) ETG – 4, and (e) ETG – 5.

crowded environment or when the matrix has a non-uniform structure causing correlated particle movement. During the time interval between 12.2 ns and 13.8 ns, the MSD curve decreases, suggesting subdiffusive behavior due to hindered diffusion caused by collisions with other particles. Finally, the curve displays a nearly linear increase in MSD value with time, representing the nearly diffusive behavior of the TFSI ion. The δ value of the MSD curve increases to 0.99. The TFSI ion is currently attempting to reach the diffusive region, leading to the eventual adoption of a diffusive behavior. Fig. S10 c. Illustrates the MSD curve depicting the motion of TFSI ions in LiTFSI of ETG – 3 over time. The MSD curve exhibits gradual motion of the TFSI ions with subdiffusive behavior. After 1.3 ns, there is an increase in the δ value from 0.24 to 0.40 until 9.3 ns? From 9.3 ns, the δ value decreases to 0.31. Fig. S10 d. illustrates the MSD curve representing the motion of TFSI ions in LiTFSI of ETG – 4 over time. Similar to ETG – 3, the TFSI ions exhibit gradual movement throughout the system until 6.1 ns, with a δ value of 0.34 indicating subdiffusive behavior. Subsequently, there is a sudden increase in the δ value to 0.71. This abrupt increase is attributed to a sudden decrease in the number of hydrogen bonds between LiTFSI and silica gel. As a result, the confinement in the system is reduced, allowing less compact space for the TFSI ion to move. Fig. S10 e. illustrates the MSD curve representing the motion of the TFSI ion in LiTFSI of ETG – 5 over time. In the initial region, the system exhibits slow movement, indicating subdiffusive behavior until 1.1 ns? The δ value of the MSD curve increases to 0.48 until 11.5 ns. This behavior is a result of the confinement of the TFSI ion within its surroundings. However, the curve experiences a sudden increase and enters the superdiffusive region. This transition is attributed to the correlated movement of

particles within the confined system. The MSD values of the TFSI anions in the eutectogels shows a consistent trend. An increase in the MSD value is observed when progressing from ETG-1 to ETG-5, indicating an increase in the diffusion of the TFSI anion. This is because the system's mean volume is nearly equal, which is given in Fig. S3 of supplementary materials. As a results, when concentration of TFSI increase, the negative charge of the TFSI ion repel each other. This is the reason that higher concentrations of TFSI anions result in greater mobility within the Eutectogel system. Surprisingly, a sudden drop in the MSD value is observed, especially with ETG-3. This unexpected behavior implies that the movement of the TFSI anions within the eutectogel is nonlinear with respect to their concentration.

Fig. S11 presents the mean square displacement (MSD) profile of the oxygen atom within the NMAC of silica-based eutectogels. In Fig. S11 a., the MSD curve of the oxygen atom in NMAC in ETG – 1 demonstrates a subdiffusive behavior, indicated by a δ value of 0.84, which signifies its confinement within a restricted region. Similarly, Fig. S11 b. shows the MSD curve of the oxygen atom in NMAC in ETG – 2, exhibiting subdiffusive behavior with a δ value of 0.85, indicating confinement to a confined region. Continuing with Fig. S11 c., the MSD curve of the oxygen atom in NMAC in ETG – 3 also reveals a subdiffusive behavior with a δ value of 0.85, indicating its confinement within a restricted region. In Fig. S11 d., the MSD curve of the oxygen atom in NMAC in ETG – 4 displays subdiffusive behavior with a δ value of 0.86, suggesting its constrained movement within a confined region. Finally, Fig. S11 e. showcases the MSD curve of the oxygen atom in NMAC in ETG – 5, where it exhibits a subdiffusive nature characterized by a δ value of 0.84, indicating confinement within a restricted region. It is important to note

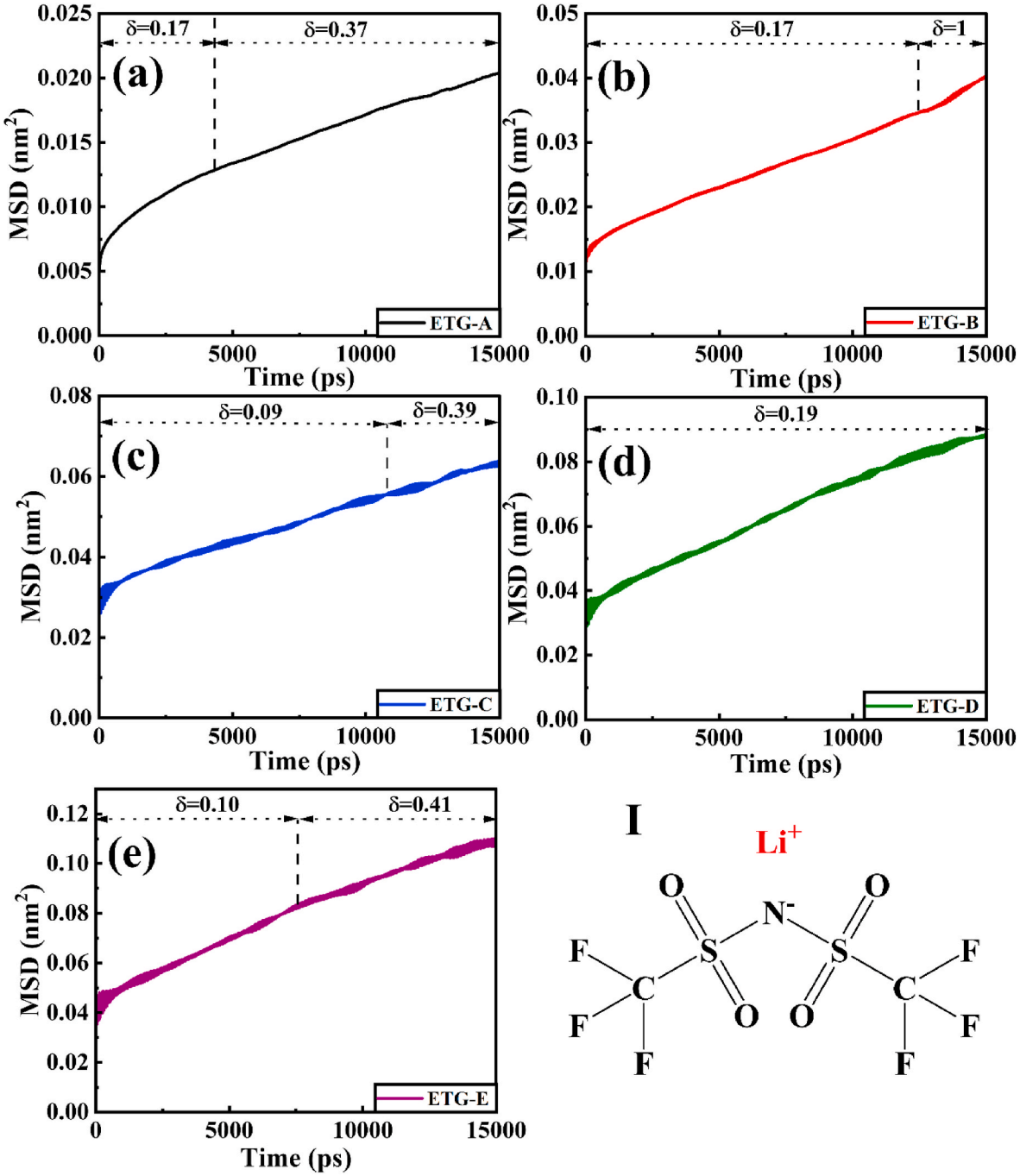


Fig. 8. The MSD curve of Li in LiTFSI in β-PVDF-HFP based eutectogel. (a) ETG – A, (b) ETG – B, (c) ETG – C, (d) ETG – D, and (e) ETG – E.

that the oxygen atom will eventually transit to diffusive behavior, but this requires a prolonged time period. The present observation shows that the MSD value of the oxygen atom in NMAC remains almost constant as the eutectogel compositions vary from ETG-1 to ETG-5. Fig. 5 a. shows that the average number of hydrogen bonds between NMAC and LiTFSI also remains relatively constant. These results suggest that the oxygen atoms are confined within their local environments, hindering their mobility and resulting in a nearly constant MSD value.

Fig. 8 illustrates the MSD profile of the lithium ion in LiTFSI of β-PVDF-HFP-based eutectogels. In Fig. 8 a., the MSD curve of the lithium ion in LiTFSI in ETG – A displays subdiffusive behavior throughout the observed time period, with two distinct regions. The first region, which lasts until 4.4 ns, exhibits a δ value of 0.17, indicating confinement within a restricted area. Beyond 4.4 ns, the δ value increases to 0.37, signifying a transition of the lithium ion from a more confined to a less confined environment. Fig. 8 b. depicts the MSD curve of the lithium ion in LiTFSI in NMAC of ETG – B. The lithium ion within the system undergoes a transition from subdiffusive to diffusive behavior over the entire time period, with the curve exhibiting two distinct regions. In the initial region, spanning until 13.9 ns, the lithium ion shows a δ value of 0.17, indicating confinement within a restricted region. However, beyond 13.9 ns, the δ value increases to 1, signifying a transition of the lithium ion from a more confined state to a progressively less confined environment. The self-diffusivity value is 4.5×10^{-13} m^2 s^{-1}. Fig. 8 c. represents the MSD curve of the lithium ion in LiTFSI in ETG – C. The lithium ion exhibits subdiffusive behavior throughout the time period, with the curve divided into two regions. The first region has a δ value of 0.09 until 10.8 ns, indicating restricted movement within a confined region. The δ value increases to 0.39, indicating a transition of the lithium ion from a more confined to a less confined environment. Fig. 8 d. represents the MSD curve of the lithium ion in LiTFSI in ETG – D. The lithium ion exhibits subdiffusive behavior throughout the time period, with a δ value of 0.19, indicating its movement is restricted within a confined region. Fig. 8 e. represents the MSD curve of the lithium ion in LiTFSI in ETG – E. The lithium ion exhibits subdiffusive behavior throughout the time period, with the curve divided into two regions. The first region has a δ value of 0.10 until 7.5 ns, indicating restricted movement within a confined region. The δ value increases to 0.41, indicating a transition of the lithium ion from a more confined to a less confined environment. Eventually, the lithium ion will exhibit diffusive behavior after a prolonged time period. The MSD value of lithium ions increases from ETG – A to ETG – E, despite the decreasing concentration of LiTFSI. This can be attributed to the collisions between the electropositive lithium atoms in LiTFSI and the fluorine atoms present in β-PVDF-HFP. The addition of β-PVDF-HFP introduces more fluorine atoms that create attractive forces with lithium ions. Consequently, lithium ions frequently collide with fluorine atoms, enhancing their diffusion and leading to the observed increase in MSD.

Fig. S12 illustrates the MSD profile of the TFSI ion in LiTFSI of β-PVDF-HFP-based eutectogels. In Fig. S11 a., the MSD curve of the TFSI ion in LiTFSI in ETG – A displays subdiffusive behavior throughout the observed time period, with two distinct regions. The first region, which lasts until 3.5 ns, exhibits a δ value of 0.19, indicating confinement within a restricted area. Beyond 3.5 ns, the δ value increases to 0.27, signifying a transition of the TFSI ion from a more confined to a less confined environment. Fig. S12 b. depicts the MSD curve of the TFSI ion in LiTFSI in NMAC of ETG – B. The TFSI ion within the system exhibits subdiffusive behavior over the entire time period, with the curve exhibiting two distinct regions. In the initial region, spanning until 13.4 ns, the TFSI ion shows a δ value of 0.18, indicating confinement within a restricted region. However, beyond 13.4 ns, the δ value increases to 0.91, signifying a transition of the TFSI ion from a more confined state to a progressively less confined environment. Fig. S12 c. Represents the MSD curve of the TFSI ion in LiTFSI in ETG – C. The TFSI ion exhibits subdiffusive behavior throughout the time period, but the curve is divided into two regions. The first region has a δ value of 0.09 until 3.8 ns, indicating restricted movement within a confined region. The δ value increases to 0.31, indicating a transition of the TFSI ion from a more confined to a less confined environment. Fig. S12 d. represents the MSD curve of the TFSI ion in LiTFSI in ETG – D. The TFSI ion exhibits subdiffusive behavior throughout the time period, but the curve is divided into two regions. The first region has a δ value of 0.07 until 1.4 ns, indicating restricted movement within a confined region. The δ value increases to 0.31 after 1.4 ns, signifying a transition of the TFSI ion from a more confined to a less confined environment. Fig. S12 e. represents the MSD curve of the TFSI ion in LiTFSI in ETG – E. The TFSI ion exhibits subdiffusive behavior throughout the time period, but the curve is divided into two regions. The first region has a δ value of 0.07 until 1.4 ns, indicating restricted movement within a confined region. The δ value increases to 0.33, indicating a transition of the TFSI ion from a more confined to a less confined environment. Eventually, the TFSI ion will exhibit diffusive behavior after an extended time period. Similar to lithium ions, the MSD values of TFSI anions exhibit an increasing trend from ETG – A to ETG – E, despite the decreasing LiTFSI concentration. This is attributed to significant collisions between TFSI and β-PVDF-HFP molecules. The presence of β-PVDF-HFP introduces additional molecular entities, increasing the likelihood of collisions with TFSI anions. These collisions enhance the diffusion of TFSI anions, resulting in the observed increase in MSD values.

Fig. S13 illustrates the mean square displacement (MSD) profile of the oxygen atom within the NMAC of β-PVDF-HFP-based eutectogels. In Fig. S13 a., the MSD curve of the oxygen atom in NMAC in ETG – A demonstrates a subdiffusive behavior with a δ value of 0.58, indicating its confinement within a restricted region. Similarly, Fig. S13 b. shows the MSD curve of the oxygen atom in NMAC in ETG – B, exhibiting subdiffusive behavior with a δ value of 0.49, suggesting its constrained movement within a confined region. Continuing with Fig. S13 c., the MSD curve of the oxygen atom in NMAC in ETG – C also reveals subdiffusive behavior with a δ value of 0.52, indicating its confinement within a restricted region. In Fig. S13 d., the MSD curve of the oxygen atom in NMAC in ETG – D displays subdiffusive behavior with a δ value of 0.56, suggesting its constrained movement within a confined region. Finally, Fig. S13 e. showcases the MSD curve of the oxygen atom in NMAC in ETG – E, where it exhibits a subdiffusive nature characterized by a δ value of 0.52, suggesting its motion is confined to a restricted region. It is important to note that the oxygen atom will eventually transition to diffusive behavior, but this requires a prolonged time period. The observation indicates that the MSD value initially decreases as the concentration of NMAC decreases. However, subsequently, it starts increasing with a further decrease in NMAC concentration. This behavior can be attributed to the strong interaction between the lithium atom in LiTFSI and the oxygen atom in NMAC. The interaction between these atoms influences the MSD value of the oxygen atom, resulting in an increasing trend.

Fig. S14 illustrates the MSD profile of the fluorine atoms attached to a single carbon atom in β-PVDF-HFP of β-PVDF-HFP-based eutectogels. In Fig. S14 a., the MSD curve of the fluorine atom in β-PVDF-HFP in ETG – A demonstrates subdiffusive behavior throughout the observed time period, characterized by a curve divided into two distinct regions. The first region, continuing until 4.5 ns, displays a δ value of 0.22, indicating confinement within a restricted region. Beyond 4.5 ns, the δ value increases to 0.42, signifying a transition of the fluorine atom from a more confined to a less confined environment. Fig. S14 b. represents the MSD curve of the fluorine atom in β-PVDF-HFP in ETG – B. The fluorine atom within the system exhibits subdiffusive behavior over the entire time period, with the curve exhibiting three distinct regions. In the initial region, spanning until 3.3 ns, the fluorine atom shows a δ value of 0.15, indicating confinement within a restricted region. However, beyond 3.3 ns, the δ value increases to 0.37, signifying a transition of the fluorine atom from a more confined state to a progressively less confined environment until 13.3 ns? Furthermore, the δ value increases to 0.82. Fig. S14 c. Represents the MSD curve of the fluorine atom in β-PVDF-

HFP in ETG – C. The fluorine atom demonstrates subdiffusive behavior throughout the time period, but the curve is divided into two regions. The first region has a δ value of 0.17 until 8.6 ns, indicating its movement is restricted within a confined region. The δ value increases to 0.42 after 8.6 ns, indicating a transition of the fluorine atom from a more confined to a less confined environment. Fig. S14 d. represents the MSD curve of the fluorine atom in β-PVDF-HFP in ETG – D. The fluorine atom displays subdiffusive behavior throughout the time period, but the curve is divided into two regions. The first region has a δ value of 0.18 until 6.9 ns, indicating its movement is restricted within a confined region. The δ value increases to 0.47 after 6.9 ns, indicating a transition of the fluorine atom from a more confined to a less confined environment. Fig. S14 e. represents the MSD curve of the fluorine atom in β-PVDF-HFP in ETG – E. The fluorine atom exhibits subdiffusive behavior throughout the time period, but the curve is divided into two regions. The first region has a δ value of 0.19 until 8.6 ns, indicating its movement is restricted within a confined region. The δ value increases to 0.49, indicating a transition of the fluorine atom from a more confined to a less confined environment. Eventually, the fluorine atoms will transition to a diffusive state after an extended period of time. The observation indicates that the MSD value of the fluorine atom increases from ETG-A to ETG-E, with the rising concentration of β-PVDF-HFP. As the β-PVDF-HFP concentration increases, more fluorine atoms become available for interactions and collisions. These interactions and collisions facilitate the enhanced diffusion of fluorine atoms, leading to the observed increase in the MSD value.

3.4. Ionic conductivity

Ionic conductivity plays a crucial role in the selection of DES for the role of electrolyte. DES displays higher viscosity as compared to traditional solvents [108] and hence displays lower ionic conductivity, i.e. less than 2 mS cm^{-1} [25]. Ionic conductivity is observed in the presence of LiTFSI in both systems. In ETG-1, the silica gel-based eutectogel, three distinct regions can be identified based on the change in the δ value. From 10 ps to 2900 ps, the σ_{GK} measures 0.012 mS cm^{-1}, which increases to 0.104 mS cm^{-1} from 2900 ps to 11560 ps. This increase can be attributed to a transition from a more confined region to a less confined one. However, a decrease in σ_{GK} is observed from 11560 ps to 15000 ps, accompanied by a decrease in the δ value. The conductivity in this region ($\sigma_{GK} = 0.016$ mS cm^{-1}) suggests the presence of obstacles impeding the movement of LiTFSI molecules. In ETG-2, the system exhibits four regions of ionic conductivity. In the first part, from 10 ps to 7560 ps , indicating subdiffusive behavior of Li ions, the σ_{GK} measures 0.082 mS cm^{-1}. From 7560 ps to 12230 ps, there is a sudden increase in the δ value, causing Li ions to enter a superdiffusive regime. However, the σ_{GK} decreases to 0.032 mS cm^{-1} within this timeframe. Subsequently, from 12230 ps to 13750 ps, Li ions transition back to a subdiffusive region as the δ value decreases from 1.15 to 0.4. The σ_{GK} during this period measures 0.03 mS cm^{-1}, attributed to the confinement of atoms hindering ion movement. Upon achieving a diffusive region beyond 13750 ps, the σ_{GK} decreases to 0.002 mS cm^{-1}. This decrease can be explained by the reduced jump distance experienced by Li ions in the superdiffusive region compared to the subdiffusive region. In ETG-3, the system can be divided into two parts based on changes in the δ value, indicating subdiffusive behavior. In the first part, the σ_{GK} measures 0.113 mS cm^{-1} until 8850 ps, while in the second part, despite changes in the δ value from 0.31 to 0.38, the σ_{GK} decreases to 0.052 mS cm^{-1}. This decrease is likely due to the confinement of Li ions, leading to smaller jumps and hindered ion movement. Similarly, in ETG-4, with subdiffusive behavior and changes in the δ value from 0.37 to 0.79, the σ_{GK} measures 0.053 mS cm^{-1} from 10 ps to 6090 ps. As Li ions transition to a less confined region, the σ_{GK} increases to 0.111 mS cm^{-1}. Finally, in the first region of the ETG-5 curve, lasting until 11520 ps, Li ions exhibit subdiffusive behavior with the σ_{GK} of 0.184 mS cm^{-1}. However, as the transition to the superdiffusive region occurs, the σ_{GK} decreases to 0.017 mS cm^{-1}. This decrease can be attributed to the reduced jump distance experienced by Li ions in the superdiffusive region compared to the subdiffusive region. Ionic conductivity of LiTFSI in silica gel based eutectogel at different timeframes are tabulated in Table 2.

In the β-PVDF-HFP based eutectogel, ETG-A exhibits subdiffusive behavior in two parts. In the first part, the σ_{GK} measures 0.015 mS cm^{-1}. With an increase in the δ value, indicating decreased confinement, the σ_{GK} increases to 0.085 mS cm^{-1}. ETG-B displays two parts as well, with a σ_{GK} of 0.093 mS cm^{-1} in the first part. While Li ions initially show subdiffusive behavior with a δ value of 0.17, a transition to diffusive behavior occurs at 13980 ps, resulting in a decreased σ_{GK} of 4.8×10^{-4} mS cm^{-1}. This decrease can be attributed to smaller jump distances by Li ions in the diffusive region compared to the subdiffusive region. In ETG-C, Li ions reside in the subdiffusive region. The σ_{GK} is 0.041 mS cm^{-1} in the first part with a δ value of 0.09. However, after 10790 ps, the σ_{GK} decreases to 0.006 mS cm^{-1} due to small jumps by Li ions in the confined region. ETG-D maintains a σ_{GK} of 0.064 mS cm^{-1} throughout the entire time frame (15000 ps) with a δ value of 0.19. Notably, in ETG-E, despite a change in the δ value from 0.10 to 0.41, the σ_{GK} remains constant at 0.013 mS cm^{-1}. This suggests that favorable ion-solvent interactions within the eutectogel environment compensate for structural variations, ensuring consistent conductivity. The ionic conductivity of LiTFSI in β – PVDF-HFP based eutectogel at different timeframes are tabulated in Table 3.

The experimental ionic conductivity value for the same system has been determined. The ionic conductivity value for the silica gel-based entactogel system is 1.46 mS cm^{-1} [35], and for the β-PVDF-HFP system is 0.30 mS cm^{-1} [109]. According to experimental literature, the ionic conductivity values of silica gel-based eutectogels are higher compared to the β-PVDF-HFP system. In the silica gel system, the lithium-ion has interacted with different moieties of TFSI ion and oxygen atoms of the silica matrix, whereas in the *β-PVDF-HFP* system, the lithium-ion only interacted with a different moiety of the TFSI ion. In the silica gel-based eutectogel system, the interacting force of lithium ions cancels each other. As a result, the mobility of lithium ions is higher, which suggests the MSD results. It is also observed in ionic conductivity results in simulation. The ionic conductivity of this paper has been calculated using the Green-Kubo relationship. The Green-Kubo method is known to be noisy, and the breakdown of correlations into atomic-level sub-correlations can pose challenges, potentially impacting the accuracy of the results [110–112]. As a result, this simulated ionic conductivity value doesn't match with the experimental ionic conductivity value.

4. Conclusion

The structural orientation and non-bonded interactions within DES systems and their interaction with organic/inorganic matrices are

Table 2
Ionic conductivity of LiTFSI in silica gel based eutectogel.

Eutectogel	Time step (ps)	Ionic conductivity (mS.cm^{-1})
ETG-1	10 → 2900	0.012
	2900 → 11560	0.104
	11560 → 15000	0.016
ETG-2	10 → 7560	0.082
	7560 → 12230	0.032
	12230 → 13750	0.003
	13750 → 15000	0.002
ETG-3	10 → 8850	0.113
	8850 → 15000	0.052
ETG-4	10 → 6090	0.053
	6090 → 15000	0.111
ETG-5	10 → 11520	0.184
	11520 → 15000	0.017

Table 3
Ionic conductivity of LiTFSI in β – PVDF-HFP based eutectogel.

Eutectogel	Time step (ps)	Ionic conductivity ($mS.cm^{-1}$)
ETG-A	20 → 4400	0.015
	4400 → 15000	0.085
ETG-B	20 → 13980	0.093
	13980 → 15000	4.8×10^{-4}
ETG-C	20 → 10790	0.041
	10790 → 15000	0.006
ETG-D	20 → 15000	0.064
ETG-E	20 → 7450	0.013
	7450 → 15000	0.013

investigated and analyzed in this study. Valuable insights into the spatial arrangement of the system and its influence on the properties of the DES-matrix composite are obtained, contributing to a deeper understanding of these materials. The hydrogen bonding interactions play a crucial role in maintaining the structural integrity and behavior of the electrolyte. The formation of DES is attributed to the weakening of the intra-molecular hydrogen bond in NMAC due to the interaction between Li^+ ions and the carbonyl group in NMAC. The number of hydrogen bonds formed between NMAC and LiTFSI remains constant in silica gel-based eutectogels, with a marginal decrease observed in the molar ratio of silica gel. Hydroxyl groups on the surface of silica gel contribute to the formation of hydrogen bonds. In contrast, β-PVDF-HFP-based eutectogels do not form hydrogen bonds between NMAC and β-PVDF-HFP due to the distance between hydrogen and fluorine atoms exceeding the threshold. The presence of hydrogen bonds influences the adhesion and wetting of the electrolyte on the electrode surface.

The RDF analysis provides insights into the interaction between different moieties of the HBA and HBD of DES, as well as their interaction with the silica gel matrix. The RDF values indicate the presence of solvation shells and the strength of interactions between different atoms. In silica gel-based eutectogels, an increase in DES concentration leads to a consistent RDF value, while a significant decrease is observed at higher concentrations, indicating increased interaction of lithium ions with other molecules. The oxygen atom of TFSI plays a crucial role, strongly interacting with lithium ions and silica atoms. In β-PVDF-HFP-based eutectogels, the RDF values increase with decreasing DES concentration, while the coordination number values remain nearly constant. The RDF curves between lithium ions and fluorine atoms within TFSI show two successive solvation shells due to strong electrostatic interaction. These results highlight the structural properties and interactions within the eutectogels, emphasizing the influence of concentration on these properties.

The dynamic behavior of lithium ions, TFSI ions, and oxygen atoms in different eutectogel compositions are investigated through MSD analysis. The MSD curves exhibit subdiffusive behavior, indicating confinement within restricted regions. The rigidity of the eutectogels, enhanced by increased hydrogen bonding with silica gel, hinders the movement of lithium ions, TFSI ions, and oxygen atoms. However, a transition from more confined to less confined environments is observed at a certain point, suggesting that over a prolonged period, diffusive behavior is expected. The addition of β-PVDF-HFP decreases the diffusion of lithium ions and TFSI ions. According to the RDF results, the Li has more interacted with the oxygen of TFSI and doesn't interact with the β-PVDF-HFP polymer in the β-PVDF-HFP system. As a result, the TFSI ion reduces the mobility of the Li-ion in the β-PVDF-HFP system. These findings demonstrate the significance of the confinement and rigidity of eutectogels in influencing the dynamic behavior of ions.

The ionic conductivity of LiTFSI in both silica gel-based eutectogels and β-PVDF-HFP based eutectogels is observed and analyzed. Silica gel-based and β-PVDF-HFP based eutectogels show distinct regions of ionic conductivity in the presence of LiTFSI. Silica gel eutectogels exhibit transitions from confined to less confined regions, affecting conductivity. β-PVDF-HFP eutectogels display subdiffusive behavior with intermittent diffusive regions. The conductivity varies due to changes in confinement and jump distances. Favorable ion-solvent interactions maintain consistent conductivity in one case.

CrediT authorship contribution statement

Shreyash P. Deshmukh: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation. **Rontu Das:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Debashis Kundu:** Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2024.234270.

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